



RAFFLES INSTITUTION

2018 YEAR 6 PRELIMINARY EXAMINATION

MATHEMATICS

9758/01

Paper 1

Total Marks: 100

3 hours

Additional materials: Answer Paper
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **9** printed pages.

RAFFLES INSTITUTION
Mathematics Department

- 1 Adult tickets for a parade are sold at three different prices, depending on the type of seats.

Children under the age of 12 and Senior Citizens aged 65 and above can enjoy 20% and 10% off the adult ticket prices respectively. Those who are between 12 and 64 years old (inclusive) will have to pay the full adult ticket price.

The number of tickets sold in each category for each group of people, together with the total cost of the tickets for each group, are given in the following table.

	Category 1 tickets (premier)	Category 2 tickets (sheltered)	Category 3 tickets (unsheltered)	Total cost
Adults (12 to 64 years old (inclusive))	150	120	200	\$18 000
Children (under 12 years old)	80	40	100	\$6 760
Senior Citizens (65 years old and above)	100	100	100	\$10 665

Write down and solve equations to find the price of an adult ticket for each of the ticket categories. [4]

- 2 Aden, a member of Raffles Art Club, is helping to design a graphic for the school's homecoming event. The main body of the graphic consists of N concentric circles $C_1, C_2, C_3, \dots, C_N$ with radii $r, 2r, 3r, \dots, Nr$, where r is a constant and N is an even integer, as shown in Fig. 1.

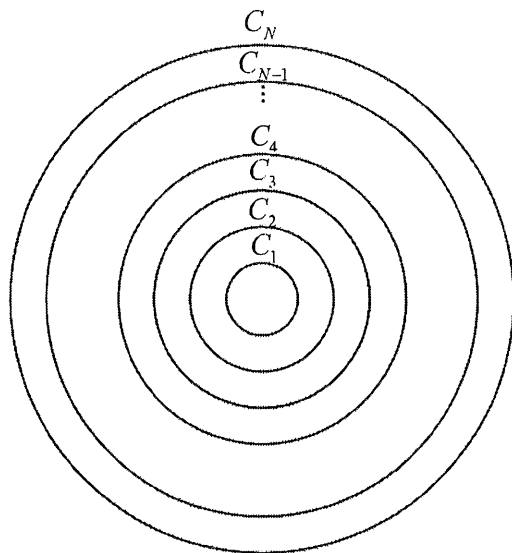


Fig. 1

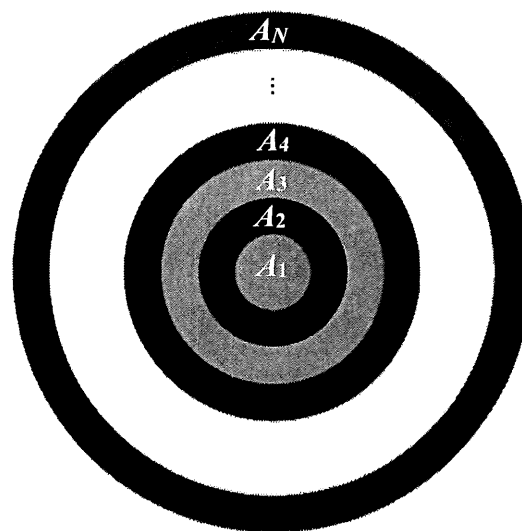


Fig. 2

The region enclosed by the circle C_1 is denoted by A_1 with area a_1 and the region between the circles C_{n-1} and C_n is denoted by A_n with area a_n for $n=2,3,4,\dots,N$, as shown in Fig. 2.

- (i) Show that the sequence $a_1, a_2, a_3, \dots, a_N$ is an arithmetic progression. [4]

With the help of a graphic software, Aden fills $A_1, A_2, A_3, \dots, A_N$ with two of the school colours: green and black. A_n will be filled green if n is odd and will be filled black if n is even.

Aden wishes to create a better visual effect by having different intensities of green. He fills A_1 with the green colour that is the same as the school colour. He reduces the intensity of this green colour by ten percent to fill A_3 ; and reduces the intensity of the green colour used to fill A_3 by ten percent to fill A_5 ; and this process continues for all odd values of n . When he finishes filling up all the areas, Aden finds that the intensity of the green colour that fills A_{N-1} falls below one quarter of the intensity of the school colour for the first time.

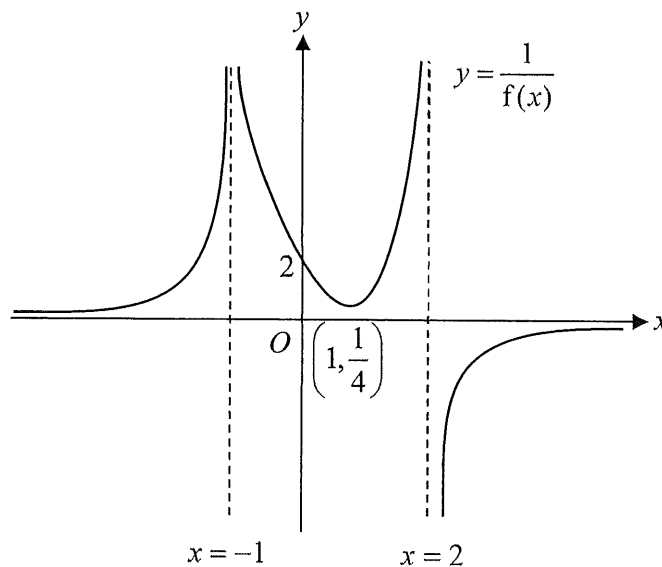
- (ii) Find the value of N . [4]

[Turn over

- 3 (a) Find $\int_0^1 \frac{9x - 4x^2}{9 - 4x^2} dx$, giving your answer in the form $p + q \ln 3 + r \ln 5$ where p, q and r are rational numbers to be determined. [5]

- (b) Prove that $\sin 3x = 3 \sin x - 4 \sin^3 x$.
Hence, or otherwise, find $\int \sin x \sin 2x \sin 3x dx$. [4]

- 4 (a) The diagram below shows the curve of $y = \frac{1}{f(x)}$ where $f(x)$ is a polynomial. The curve has a minimum point at $\left(1, \frac{1}{4}\right)$ and cuts the y -axis at $(0, 2)$. The lines $x = -1$ and $x = 2$ are the vertical asymptotes and the line $y = 0$ is the horizontal asymptote to the curve.



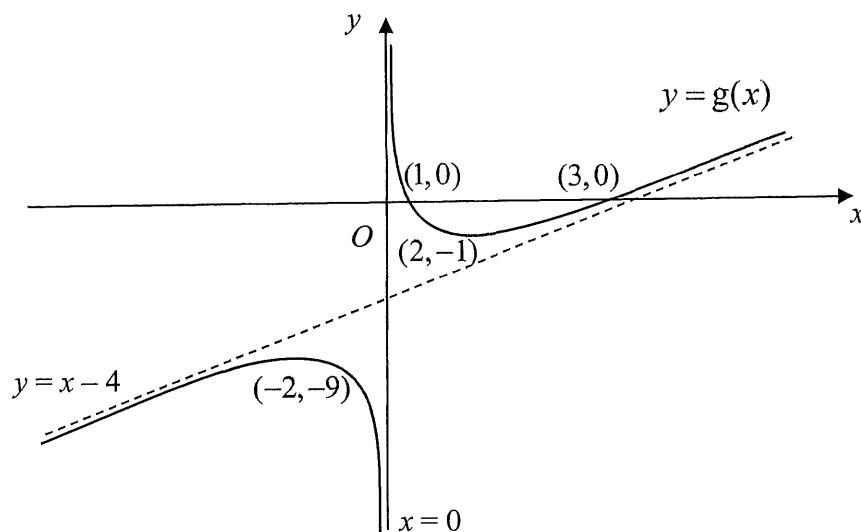
Sketch on separate diagrams, the graphs of

- (i) $y = f(x)$, [3]

- (ii) $y = f(|x|)$, [2]

labelling all relevant point(s).

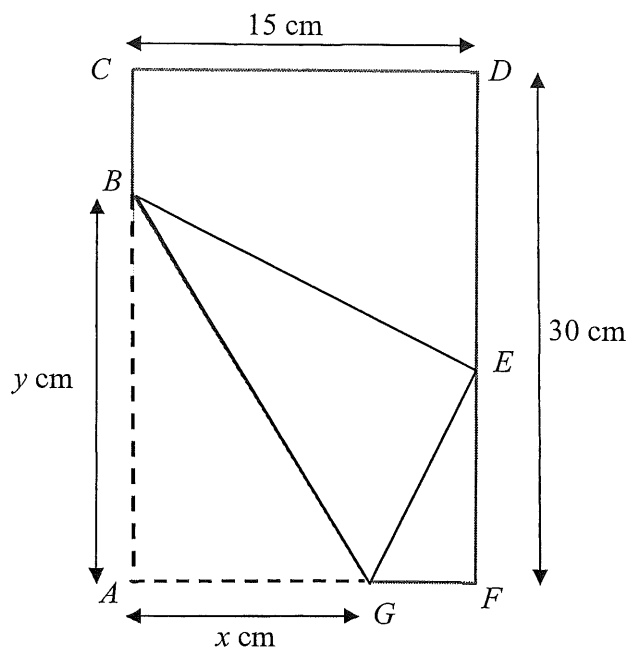
- 4 (b) The diagram below shows the curve of $y = g(x)$. The curve has a maximum point at $(-2, -9)$ and a minimum point at $(2, -1)$. The curve crosses the x -axis at $(1, 0)$ and $(3, 0)$. The line $x = 0$ is the vertical asymptote and the line $y = x - 4$ is the oblique asymptote to the curve.



- (i) Sketch the graph of $y = g'(x)$, labelling all relevant point(s) and stating the equations of any asymptotes. [2]
- (ii) Find the area bounded by the graph of $y = g'(x)$, the lines $x = 1$, $x = 2$ and the x -axis. [2]

[Turn over]

5



Tom has a rectangular piece of paper $ACDF$ with length 30 cm and breadth 15 cm. He folds the lower left-hand corner, A , to reach the rightmost edge of the paper at E . After that he will cut out the triangle EFG and the trapezium $BCDE$ to obtain a kite shaped figure $ABEG$ with AB of length y cm and AG of length x cm.

- (i) Find the length of EF in terms of x . [2]
- (ii) Show that $y = \frac{x\sqrt{15}}{\sqrt{2x-15}}$. [2]
- (iii) Using differentiation, find the exact values of x and y which give the minimum area of the kite $ABEG$. [5]

6 (a) It is given that $e^y = \tan\left(x + \frac{\pi}{3}\right)$.

(i) Show that $\frac{dy}{dx} = e^y + e^{-y}$. [2]

(ii) Using differentiation, find the Maclaurin series for $\ln\left[\tan\left(x + \frac{\pi}{3}\right)\right]$ up to and including the term in x^2 . [3]

(b) Let $f(x) = \frac{3x^2 - 4x + 5}{(1+x)(1-x)^2}$.

(i) Express $f(x)$ in the form $\frac{A}{(1+x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$ where A , B and C are constants to be determined. [2]

(ii) Hence find the expansion of $f(x)$ up to and including the term in x^4 . [2]

(iii) Write down the coefficient of x^r in the expansion of $f(x)$ in terms of r . [1]

7 Do not use a calculator in answering this question.

(a) Find the roots of the equation $w^2(1-i) + 4w + (10+10i) = 0$, giving your answers in cartesian form $a+ib$. [3]

(b) It is given that $z = -\sqrt{3} + i$.

(i) Find an exact expression for z^5 . Give your answer in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

Not in the current syllabus

(ii) Find the three smallest positive whole number values of n for which $\frac{z^n}{iz^*}$ is purely imaginary. [4]

(iii) Given that $\left|1 + \frac{p}{z^*}\right| = \sqrt{7}$, find exactly the possible values of the real number p . [3]

[Turn over]

- 8 The function $f(x) = a - (1+x)^{\frac{1}{3}}$ is defined for $0 \leq x \leq 7$, where a is a positive real constant.

- (i) Solve the inequality $f(x) \geq a - 1$. [2]
- (ii) Show that the graph of $y = f(x)$ has a negative gradient at all points on the graph. Find the range of f . [2]

Use $a = 3$ in the rest of the question.

- (iii) Find $f^{-1}(x)$. [2]
- (iv) Sketch on the same diagram the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the geometrical relationship between the two graphs and the line $y = x$. [3]
- (v) Show that the x -coordinate of the point of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $x^3 - 9x^2 + 28x - 26 = 0$. Hence find the solution of the equation $f(x) = f^{-1}(x)$. [3]

- 9 (a) The variables x and y are related by the differential equation $\frac{dy}{dx} = \sqrt{x}e^{\sqrt{x}}$. Using the substitution $u = \sqrt{x}$, show that $y = k \int u^2 e^u du$ where k is a constant to be determined. Hence find the general solution of the differential equation. [5]

- (b) Ms Frugal bought a second-hand oven which has a broken timer. Each time she bakes she uses the timer on her handphone. One day, she decides to make a loaf of bread. She takes the fermented dough out of her refrigerator and checks that the internal temperature of the dough is 4°C . She puts it into the oven which has been pre-heated to a constant temperature of 180°C and forgets to set the timer on her handphone. She also does not note the time. At 10.06 am, she realizes that the timer has not been set. She checks her dough which now has an internal temperature of 80°C . Twenty minutes later, she checks the dough again and the internal temperature has risen to 130°C .

Newton's Law of Cooling states that the rate of increase of the temperature $\theta^\circ\text{C}$ of an object after t minutes is directly proportional to the difference in the temperatures of the object and its surrounding.

The dough has to be baked at a constant oven temperature of 180°C for one and a half hour to cook through. By forming a differential equation, find the time (to the nearest minute) that Ms Frugal should remove the dough from the oven. [8]

- 10 Two children Hansel and Gretel are participating in the junior category of the Festival of Lights competition. They decided to set up a structure consisting of a pyramid with a rectangular base $OABC$ and vertex V . The main power supply switch is positioned at O with coordinates $(0, 0, 0)$ and relative to O , the coordinates of A and C are $(1, 2, 1)$ and $(-1, 1, -1)$ respectively. The vertex V is fixed at a height $\sqrt{8}$ metres above the rectangular base $OABC$ such that it is equidistant to the points O , A , B and C . All the surfaces of the pyramid, excluding the base $OABC$, are completely covered with LED (light emitting diodes) light strips so that it illuminates in the dark.

- (i) Point D is the foot of the perpendicular from V to the rectangular base $OABC$. Show that the coordinates of D is $(0, 1.5, 0)$. [1]
- (ii) Find a vector perpendicular to the rectangular base $OABC$. Hence, find the position vector of V , given that its $\mathbf{k}$ -component is positive. [4]

Hansel and Gretel decide to install two different colour display schemes on the pyramid and the control switch is to be installed inside the pyramid at a point E with position vector $\alpha\mathbf{i} + 1.5\mathbf{j} - \alpha\mathbf{k}$, where α is a constant.

- (iii) Show that VE is perpendicular to the rectangular base $OABC$ and explain why $-2 < \alpha < 0$. [3]
- (iv) Find the distance between E and the surface OVA in terms of α , simplifying your answer. [4]
- (v) Given that the ratio of the distance between E and the rectangular base $OABC$ to the distance between E and the surface OVA is $10:\sqrt{105}$, find the value of α . [2]

***** End of Paper *****